



Configuring Port Security for a Switch

Difficulty (out of 3 stars): ★★★

Recommended study time: 1 hour

Contents

1 About This Lab	2
1.1 Task Description	2
1.2 Objectives	2
2 Task Preparation	2
2.1 Network Topology	2
2.2 Initial Configuration	3
3 Task Implementation	4
3.1 Checking Basic Configurations	4
3.2 Configuring Dynamic Port Security	4
3.3 Configuring Sticky Port Security	6
4 Verification	7
5 Quiz	7

1 About This Lab

1.1 Task Description

Port security converts the dynamic MAC addresses learned by a port into secure MAC addresses (including dynamic and static secure MAC addresses, and sticky MAC addresses). This function prevents unauthorized users from using this port to communicate with the switch and therefore enhances device security.

On completion of this task, you will gain a clear understanding of port security on Huawei's switches and be able to configure port security to enhance device security.

1.2 Objectives

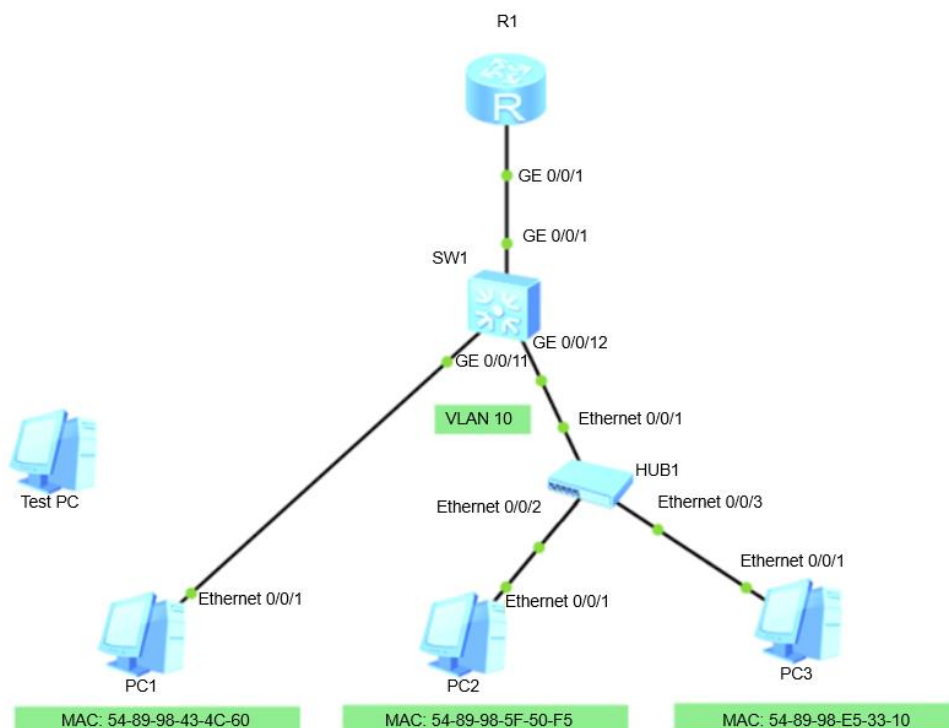
On completion of this task, you will be able to:

1. Understand the fundamentals of port security.
2. Know application scenarios of port security.
3. Configure port security.

2 Task Preparation

2.1 Network Topology

Figure 2-1 Implementing security on switch ports



2.2 Initial Configuration

- Initial configuration for R1

```
[Huawei]sysname R1
[R1]int g0/0/1
[R1-GigabitEthernet0/0/1]ip address 10.0.1.254 24
[R1-GigabitEthernet0/0/1]quit
[R1]
```

- Initial configuration for SW1

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname SW1
[SW1]
[SW1]vlan 10
[SW1-vlan10]quit
[SW1]
[SW1]int g0/0/1
[SW1-GigabitEthernet0/0/1]port link-type access
[SW1-GigabitEthernet0/0/1]port default vlan 10
[SW1-GigabitEthernet0/0/1]quit
[SW1]
[SW1]int g0/0/11
[SW1-GigabitEthernet0/0/11]port link-type access
[SW1-GigabitEthernet0/0/11]port default vlan 10
[SW1-GigabitEthernet0/0/11]quit
[SW1]
[SW1]int g0/0/12
[SW1-GigabitEthernet0/0/12]port link-type access
[SW1-GigabitEthernet0/0/12]port default vlan 10
[SW1-GigabitEthernet0/0/12]quit
[SW1]
```

- IP addresses and MAC addresses of PC1, PC2, and PC3

```
IP address of PC1: 10.0.1.11/24; MAC address of PC1: 54-89-98-43-4C-60
IP address of PC2: 10.0.1.12/24; MAC address of PC2: 54-89-98-5F-50-F5
IP address of PC3: 10.0.1.13/24; MAC address of PC3: 54-89-98-E5-33-10
```

3 Task Implementation

3.1 Checking Basic Configurations

1. On PC1, PC2, and PC3, ping the gateway IP address 10.0.1.254 to test initial network connectivity.

On PC1:

```
PC>ping 10.0.1.254 -c 2
```

```
Ping 10.0.1.254: 32 data bytes, Press Ctrl_C to break
```

```
From 10.0.1.254: bytes=32 seq=1 ttl=255 time=31 ms
```

```
From 10.0.1.254: bytes=32 seq=2 ttl=255 time=31 ms
```

On PC2:

```
PC>ping 10.0.1.254 -c 2
```

```
Ping 10.0.1.254: 32 data bytes, Press Ctrl_C to break
```

```
From 10.0.1.254: bytes=32 seq=1 ttl=255 time=47 ms
```

```
From 10.0.1.254: bytes=32 seq=2 ttl=255 time=31 ms
```

On PC3:

```
PC>ping 10.0.1.254 -c 2
```

```
Ping 10.0.1.254: 32 data bytes, Press Ctrl_C to break
```

```
From 10.0.1.254: bytes=32 seq=1 ttl=255 time=47 ms
```

```
From 10.0.1.254: bytes=32 seq=2 ttl=255 time=47 ms
```

3.2 Configuring Dynamic Port Security

1. Configure GE0/0/12 on SW1.

Deploy port security on SW1. Specify that the maximum number of MAC addresses that can be learned by GE0/0/12 is 1. If the interface is connected to multiple PCs, SW1 generates an alarm.

```
[SW1-GigabitEthernet0/0/11]int g0/0/12
```

```
[SW1-GigabitEthernet0/0/12] port-security enable
```

```
[SW1-GigabitEthernet0/0/12]port-security max-mac-num 1
```

```
[SW1-GigabitEthernet0/0/12]port-security protect-action restrict
```

```
[SW1-GigabitEthernet0/0/12]quit
```

```
[SW1]
```

Connect a hub to GE0/0/12 and connect two PCs to the hub. Run the **ping** command on PC2 and PC3 to ping 10.0.1.254. The following alarm is displayed on the switch, and the ping is successful for only one PC.

```
<SW1>
```

```
Apr 12 2025 15:23:43-08:00 SW1 L2IFPPI/4/PORTSEC_ACTION_ALARM:OID 1.3.6.1.4.1.20
11.5.25.42.2.1.7.6 The number of MAC address on interface (17/17) GigabitEthernet
0/0/12 reaches the limit, and the port status is : 1. (1:restrict;2:protect;3:s
hutdown)
```

```
<SW1>
```

```
<SW1>
```

```
Apr 12 2025 15:25:06-08:00 SW1 L2IFPPI/4/PORTSEC_ACTION_ALARM:OID 1.3.6.1.4.1.20
11.5.25.42.2.1.7.6 The number of MAC address on interface (17/17) GigabitEthernet
0/0/12 reaches the limit, and the port status is : 1. (1:restrict;2:protect;3:s
hutdown)
```

```
<SW1>
```

2. Check the dynamic secure MAC address entries on SW1.

```
<SW1>display mac-address security
```

```
MAC address table of slot 0:
```

MAC Address	VLAN/ VSI/SI	PEVLAN	CEVLAN	Port	Type	LSP/LSR-ID MAC-Tunnel
5489-98e5-3310	10	-	-	GE0/0/12	security	-

```
Total matching items on slot 0 displayed = 1
```

```
<SW1>
```

3. Specify that the maximum number of MAC addresses that can be learned by GE0/0/12 is 1. If the number of learned MAC addresses exceeds the limit, the switch generates an alarm and shuts down the interface.

```
[SW1]int g0/0/12
```

```
[SW1-GigabitEthernet0/0/12] port-security enable
```

```
[SW1-GigabitEthernet0/0/12]port-security max-mac-num 1
```

```
[SW1-GigabitEthernet0/0/12]port-security protect-action shutdown
```

```
[SW1-GigabitEthernet0/0/12]quit
```

```
[SW1]
```

4. Run the **ping** command on PC2 and PC3 to ping 10.0.1.254. The following alarm is displayed on the switch, and GE0/0/12 is shut down.

```
[SW1-GigabitEthernet0/0/12]
```

```
Apr 12 2025 15:32:31-08:00 SW1 L2IFPPI/4/PORTSEC_ACTION_ALARM:OID 1.3.6.1.4.1.20
11.5.25.42.2.1.7.6 The number of MAC address on interface (17/17) GigabitEthernet
0/0/12 reaches the limit, and the port status is : 3. (1:restrict;2:protect;3:s
hutdown)
```

```
Apr 12 2025 15:32:31-08:00 SW1 %%01PHY/1/PHY(I)[0]: GigabitEthernet0/0/12: ch
ange status to down
```

```
[SW1-GigabitEthernet0/0/12]

# Enable G0/0/12 manually.
[SW1]int g0/0/12
[SW1-GigabitEthernet0/0/12]undo shutdown
[SW1]

# Run the display mac-address security command. No command output is displayed.
[Switch1]display mac-address security
```

3.3 Configuring Sticky Port Security

1. Configure GE0/0/11 on SW1.

Deploy port security on SW1. Enable port security on GE0/0/11. Specify that the maximum number of MAC addresses that can be learned by GE0/0/11 is 1 and convert dynamic secure MAC addresses learned by GE0/0/11 into sticky MAC addresses.

```
[SW1]
[SW1]interface GigabitEthernet 0/0/11
[SW1-GigabitEthernet0/0/11]port-security enable
[SW1-GigabitEthernet0/0/11]port-security max-mac-num 1
[SW1-GigabitEthernet0/0/11] port-security mac-address sticky
[SW1-GigabitEthernet0/0/11]quit
[SW1]
```

2. Ping the gateway IP address from PC1.

```
PC>ping 10.0.1.254
```

```
Ping 10.0.1.254: 32 data bytes, Press Ctrl_C to break
From 10.0.1.254: bytes=32 seq=1 ttl=255 time=47 ms
From 10.0.1.254: bytes=32 seq=2 ttl=255 time=46 ms
From 10.0.1.254: bytes=32 seq=3 ttl=255 time=16 ms
```

3. Run the **display mac-address sticky** command on SW1.

```
[SW1]display mac-address sticky
MAC address table of slot 0:
```

MAC Address	VLAN/ VSI/SI	PEVLAN	CEVLAN	Port	Type	LSP/LSR-ID MAC-Tunnel
5489-9843-4c60	10	-	-	GE0/0/11	sticky	-

```
Total matching items on slot 0 displayed = 1
```

```
[SW1]
```

The command output shows that GE0/0/11 is bound to 5489-9843-4c60, the MAC address of PC1.

4. Connect the test PC to GE0/0/11, and ping the gateway IP address from the test PC (IP address: 10.0.1.20). The ping fails.

[SW1]

```
Apr 12 2025 15:47:17-08:00 SW1 L2IFPPI/4/PORTSEC_ACTION_ALARM:OID 1.3.6.1.4.1.2011.5.25.42.2.1.7.6 The number of MAC address on interface (16/16) GigabitEthernet0/0/11 reaches the limit, and the port status is : 1. (1:restrict;2:protect;3:shutdown)
```

4 Verification

Omitted

5 Quiz

1. What is the command that displays dynamic secure MAC address entries?
display mac-address security
2. What is the command that displays secure static MAC address entries?
display mac-address sec-config
3. What is the command that displays sticky MAC address entries?
display mac-address sticky